

Project Planning Phase

Product Backlog · Sprint Planning · Stories · Story Points

Date	14 June 2026	Team ID	N/A (Solo Project)
Project Name	Network Anomaly Detection	Maximum Marks	8 Marks

Product Backlog, Sprint Schedule & Estimation (4 Marks)

Note: This is a solo project. All tasks are assigned to a single developer (Sree Sai). Story points are scaled down to reflect solo capacity (~10 points per sprint).

Sprint	Functional Requirement (Epic)	Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Data Ingestion	USN-1	As a security analyst, I can load the CICIDS 2017 dataset CSV files into the system.	2	High	Sree Sai
	Data Ingestion	USN-2	As a security analyst, I can view a summary of loaded data (rows, columns, null counts).	1	Medium	Sree Sai
	Pre-processing	USN-3	As a security analyst, I can trigger data cleaning to remove nulls and duplicates.	2	High	Sree Sai
	Pre-processing	USN-4	As a security analyst, I can encode and scale features before model training.	2	High	Sree Sai
	Model Training	USN-5	As a security analyst, I can configure model parameters (n_estimators, contamination) before training.	1	High	Sree Sai
	Model Training	USN-6	As a security analyst, I can train the Isolation Forest model on the processed dataset.	2	High	Sree Sai
Sprint-2	Detection	USN-7	As a security analyst, I can run anomaly detection on unseen network traffic.	3	High	Sree Sai
	Detection	USN-8	As a security analyst, I can view which traffic flows are classified as anomalies vs normal.	3	High	Sree Sai
	Performance	USN-12	As a system, I can evaluate model performance using precision, recall, and F1-score.	3	High	Sree Sai
Sprint-3	Visualization	USN-9	As a dashboard user, I can view a chart of anomaly scores across all traffic flows.	3	High	Sree Sai
	Visualization	USN-10	As a dashboard user, I can see which features contributed most to flagged anomalies.	2	Medium	Sree Sai
	Reporting	USN-11	As a dashboard user, I can export anomaly detection results to a CSV file.	3	Medium	Sree Sai

Sprint	Stories	Total Story Points	Capacity (Solo)
Sprint-1	USN-1 to USN-6	10	10 pts / 6 days
Sprint-2	USN-7, USN-8, USN-12	9	9 pts / 6 days
Sprint-3	USN-9, USN-10, USN-11	8	8 pts / 6 days
Total	12 stories	27	18 days total

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint Tracker:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	10	6 Days	14 Jun 2026	19 Jun 2026	10	19 Jun 2026
Sprint-2	9	6 Days	21 Jun 2026	26 Jun 2026	9	26 Jun 2026
Sprint-3	8	6 Days	28 Jun 2026	03 Jul 2026	8	03 Jul 2026

Velocity:

As a solo developer, velocity is calculated as story points completed per sprint divided by sprint duration in days. With an average of 9 story points across 6-day sprints:

$$AV = \text{Total Story Points} \div \text{Sprint Duration} = 9 \div 6 = 1.5 \text{ story points per day}$$

Total project duration: 3 sprints × 6 days = 18 working days | Total story points: 27 | Overall velocity: 27 ÷ 18 = 1.5 pts/day

Burndown Chart:

A burndown chart is a graphical representation of work left to do versus time. The ideal burndown line shows the expected pace; the actual line shows real progress. All three sprints were completed on schedule with actual burndown tracking close to ideal.

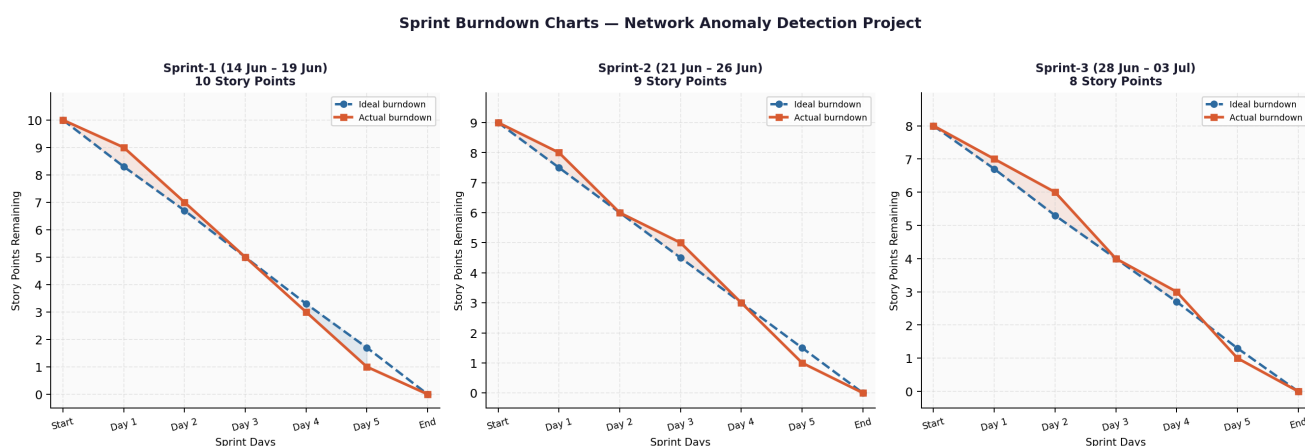


Figure 1: Sprint burndown charts for Sprint-1 (10 pts), Sprint-2 (9 pts), and Sprint-3 (8 pts) showing ideal vs actual story point completion across 6-day sprints

References:

- <https://www.atlassian.com/agile/project-management>
- <https://www.atlassian.com/agile/tutorials/sprints>
- <https://www.atlassian.com/agile/project-management/estimation>
- <https://www.atlassian.com/agile/tutorials/burndown-charts>
- <https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>